

ST CHARLES COUNTY AP, MO, USA

WMO: 724347

Lat: 38.929N

Lon: 90.428W

Elev: 133

StdP: 99.74

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 53904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.5	-20.0	0.6	-12.8	-17.6	0.8	-10.5	11.8	0.5	11.0	0.3	3.6	320	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	35.3	24.9	33.8	24.5	32.4	24.0	26.6	32.0	25.8	31.2	25.1	30.4	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.6	29.3	24.2	19.4	28.5	23.5	18.6	27.9	83.5	31.9	80.2	31.3	76.9	30.4	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.0	8.0	DB	-18.0	37.6	2.9	1.5	-20.0	38.7	-21.7	39.6	-23.3	40.4	-25.4	41.6	(4)
(5)			WB	-18.4	27.9	2.8	1.1	-20.4	28.6	-22.0	29.2	-23.6	29.8	-25.6	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-0.6	0.9	7.5	13.8	19.1	24.2	25.7	24.9	20.7	14.1	7.6	1.8
	DBStd	10.48	6.17	6.13	5.97	5.07	4.49	3.13	3.08	3.01	3.98	4.98	5.21	5.34
	HDD10.0	1113	331	259	121	21	1	0	0	0	0	16	106	258
	HDD18.3	2617	586	487	338	150	47	2	0	1	21	150	323	512
	CDD10.0	2344	3	6	42	135	283	427	486	461	320	144	33	5
	CDD18.3	807	0	0	2	14	71	179	227	203	91	19	1	0
	CDH23.3	8072	0	1	29	187	652	1778	2408	1941	867	201	9	0
	CDH26.7	3322	0	0	2	46	202	742	1094	851	331	53	1	0
Wind	WSAvg	3.5	4.0	4.1	4.3	4.5	3.6	2.9	2.4	2.3	2.6	3.4	3.7	3.8
Precipitation	PrecAvg	1057	62	57	80	116	126	113	96	86	83	82	89	66
	PrecMax	1475	214	127	175	279	281	293	202	154	259	332	188	281
	PrecMin	740	18	11	29	47	26	14	19	23	11	16	17	11
	PrecStd	201	42	30	33	55	67	67	48	35	54	60	48	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	21.2	26.4	30.7	32.8	35.8	38.4	37.2	34.5	31.0	24.4	19.1
		MCWB	13.9	14.1	17.1	19.9	22.6	24.1	24.5	25.0	23.3	22.2	17.2	15.5
	2%	DB	14.8	17.2	23.3	27.8	30.8	34.0	35.6	35.0	32.5	27.9	21.4	15.3
		MCWB	11.1	12.3	16.0	18.2	21.9	24.2	25.4	24.7	23.2	20.1	15.3	12.1
	5%	DB	11.5	13.6	20.1	25.5	28.9	32.7	33.7	33.0	30.6	25.7	18.7	12.5
		MCWB	8.2	9.9	14.3	16.9	20.9	24.0	25.2	24.2	22.1	18.4	14.5	9.8
	10%	DB	8.2	10.2	17.4	22.9	27.3	31.1	32.1	31.3	28.2	22.8	16.3	9.7
		MCWB	5.7	7.3	13.1	16.1	20.2	23.1	24.5	23.7	20.8	17.4	12.3	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	16.0	18.9	21.7	24.5	26.7	27.9	27.4	25.2	23.0	18.9	16.4
		MCDB	18.5	19.7	24.6	27.9	30.2	32.1	33.1	33.0	31.3	28.6	22.5	18.4
	2%	WB	12.0	13.3	16.9	19.6	23.0	25.7	26.8	26.3	24.3	21.7	16.9	13.2
		MCDB	14.3	16.3	21.5	25.1	28.3	31.6	32.1	31.8	29.8	26.3	19.8	15.1
	5%	WB	8.6	10.2	15.5	18.3	22.1	24.9	26.1	25.4	23.5	20.2	14.9	10.2
		MCDB	11.0	13.3	19.3	23.5	27.2	30.6	31.3	30.8	28.0	23.6	18.0	12.0
	10%	WB	5.5	7.0	13.6	17.0	21.1	24.0	25.3	24.6	22.5	18.3	12.8	7.8
		MCDB	8.4	10.1	16.7	21.8	25.9	29.2	30.5	29.6	26.4	21.9	16.0	9.3
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.9	11.2	11.9	11.0	10.9	10.6	11.2	12.3	11.9	10.9	8.7
		MCDBR	15.2	15.9	15.7	16.1	13.1	12.7	12.4	13.4	14.1	15.1	15.5	13.1
		MCWBR	11.2	11.4	9.5	8.3	5.9	5.0	4.5	5.0	5.5	7.6	10.1	10.1
	5% WB	MCDBR	14.1	14.5	13.7	13.5	11.3	11.2	11.0	11.7	11.7	12.1	12.9	11.8
	MCWBR	11.2	11.2	9.1	8.1	5.9	5.1	4.9	5.1	5.2	7.2	9.6	9.9	
Clear-Sky Solar Irradiance	taub	0.304	0.315	0.346	0.401	0.440	0.446	0.476	0.464	0.411	0.353	0.310	0.300	
	taud	2.467	2.428	2.410	2.253	2.205	2.226	2.141	2.175	2.293	2.463	2.526	2.494	
	Ebn at Noon	874	911	912	875	845	837	808	807	831	856	862	851	
	Edh at Noon	80	95	107	134	143	140	151	143	119	91	75	73	
All-Sky Solar Radiation	RadAvg	2.02	2.75	3.73	4.92	5.54	6.33	6.22	5.54	4.75	3.38	2.29	1.70	
	RadStd	0.23	0.27	0.34	0.29	0.47	0.50	0.42	0.32	0.30	0.37	0.23	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82

Nomenclature: See separate page